

HR Attrition Analysis & Prediction

Business report | IBM HR Analytics dataset (1,470 employees) | Tools: MySQL, Python (scikit-learn), Power BI

1,470

Total employees

237

Employees who left

16.1%

Attrition rate

\$6,503

Avg monthly income

212

Predicted to leave

17.19%

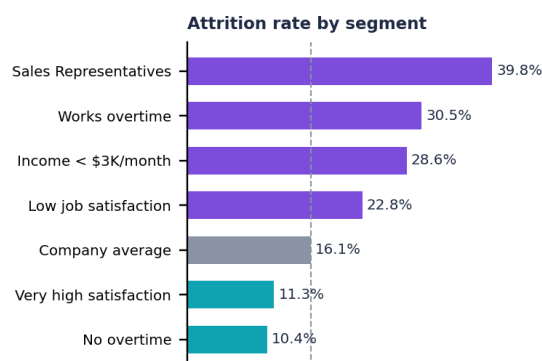
Predicted risk rate

1. Business Problem

Employee attrition is costly: every leaver means recruitment, onboarding and lost productivity. In this dataset **237 of 1,470 employees (16.1%) have left**. Leadership needs to know **who is leaving, why, and who is likely to leave next** so that retention money is spent where it matters. This project answers three questions: (1) which groups have the highest attrition, (2) is pay a statistically significant factor, and (3) which of the 1,233 current employees are at risk.

2. Key Insights (Descriptive Analysis)

- Overtime is the strongest driver.** 30.5% of overtime workers left vs 10.4% of others, almost **3x higher**.
- Role matters.** Sales Representatives have the highest attrition rate (39.8%), then Laboratory Technicians (23.9%) and HR staff (23.1%). Lab Technicians also have the most leavers in absolute terms (62).
- Department:** Sales 20.6%, Human Resources 19.1%, Research & Development 13.8%.
- Low pay, high exit.** Employees earning under \$3K a month leave at 28.6%, versus 8.9% for those above \$10K.
- Satisfaction:** employees with the lowest job satisfaction leave at 22.8%, about **2x** the rate of the most satisfied (11.3%).
- Gender:** 63% of leavers are male (150) and 37% female (87), broadly in line with the 60/40 workforce mix, so gender is not a driver.



3. Hypothesis Test: Does Income Differ Between Leavers and Stayers?

Item	Detail
H_0 / H_1	H_0 : mean monthly income is the same for employees who stayed and who left. H_1 : the means are different.
Test	Independent two-sample t-test (Welch, unequal variances), alpha = 0.05
Evidence	Average income: \$6,833 (stayed) vs \$4,787 (left), a gap of about \$2,046 (30% lower for leavers). $t = 7.48$, $p = 4.4 \times 10^{-13}$
Decision	Reject H_0 . Income is significantly lower among leavers.

Interpretation: the result is statistically significant, but it shows association, not proof of cause. Income is also tied to job level and tenure, so pay should be treated as one factor among several.

4. Machine Learning Prediction

Approach. Identifiers and gender were excluded, categorical fields were one-hot encoded, numeric fields were standardised, and data was split 80/20 (random_state = 42). Two models were tuned with 5-fold GridSearchCV:

- Random Forest:** 87% accuracy but only **10% recall** on leavers (caught 4 of 39). Accuracy looks good only because 84% of employees stay, so it was rejected.
- Logistic Regression (L1, C = 1):** selected. At the default 0.50 cut-off, recall was 36%. Lowering the cut-off to **0.40** gave the best balance: **precision 0.50, recall 0.49, F1 0.49** for leavers, 87% overall accuracy.

A 0.40 threshold was chosen because missing a real leaver costs more than an unnecessary check-in.

Results on the 1,233 current employees: **212 (17.19%)** are flagged as likely to leave; average attrition probability is 21.3%. By department: R&D 123, Sales 77, HR 12. By role: Laboratory Technician 60, Sales Executive 49, Research Scientist 36, Sales Representative 20. **140 of the 212 (66%) work overtime**, which matches the overtime finding above.

5. Business Recommendations

- Control overtime first.** Cap or rebalance overtime, starting with Laboratory Technicians and Sales teams. Managers should hold a check-in with any overtime worker whose risk score is above 40%.
- Review pay for low-income, entry-level staff.** Benchmark roles paid under \$3K a month and apply targeted pay or bonus adjustments rather than across-the-board raises.
- Fix the Sales Representative and Lab Technician roles.** Sales Representatives (39.8%) need clearer targets, career paths and onboarding support; Lab Technicians need workload relief and a promotion route.
- Act on satisfaction scores.** Run quarterly pulse surveys and follow up within 30 days on any employee scoring 'Low' job satisfaction.

- **Operationalise the model.** Review the flagged list monthly, rank by probability, and run stay interviews with the highest-risk employees. Track how many flagged employees actually leave and retrain the model each quarter.

Limitations: the IBM HR dataset is a sample dataset and a snapshot, so results should be validated on real company data. With recall near 49%, the model is a prioritisation tool for conversations, not a verdict on any individual. The 1,233 scored employees overlap with the training data, and the 0.40 cut-off was picked using the same test split, so live performance may be somewhat lower.

Appendix: Power BI Dashboards



Page 1: Attrition Overview

Answers *what happened*: total attrition, and rates by gender, job role, department, income band and job satisfaction.



Page 2: Predicted Attrition Risk

Answers *what may happen next*: 212 flagged employees by role, department and overtime, with a probability table (flag = probability above 40%).